# Examination –Dec- 2022

# B.Tech. I Sem : CSE, IT, AIADS, CSE(BC), IoT

# Digital Electronics

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

**Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.**

**Word limit be observed as follows:**

**Part a – Max 50 words, Part b – Max 50 words,**
**Part c – Max 100 words and Part d – Max 400 words.**

**Word limit NOT to be followed for diagram, numerical, derivation.**

Q.1 (a) Add $(756)_8 + (ACC)_{16}$ 02

(b) Representation of 7 in 8 4 -2 -1 code 02

(c) Perform the BCD addition $(175)_{10} + (259)_{10}$ 03

**OR**

Convert $(135)_{10}$ to Gray code 03

(d) A 7-bit Hamming code 1010110 is received in which almost a single error has occurred. Locate the position of the error bit using parity checks assuming code was created using an even parity 05

**OR**

Solve the following : 05

(a) Convert $(01FA.3A)_{16}$ to $(N)_2$

(b) $(255)_{10}$ to $(N)_8$

(c) Subtract $(011011)_2 - (110011)_2$

(d) $(1721.54)_8$ to $(N)_{10}$

(e) 2's complement of $(1011010)_2$

Q.2 (a) Obtain the simplified expressions in sum of products for the following Boolean functions: $F(x,y,z)= \sum(2,3,12,13,14,15)$ 02

(b) $F= \pi (0,1,2,3,4,10,11)$, find POS. 02

(c) Simplify the Boolean functions F using the don't-care conditions in Sum of product form 03

F= A'B'D'+ A'CD+ A'BC

d= A'BC'D +ACD +AB'D'

**OR**

Find POS 03

F = (A+B'+D) (A'+B+D) (C+D) (C'+D')

Scanned with OKEN Scanner

(d) Implement the function using NAND gate in SOP form 04

F= B'D + B'C +ABCD

d = A'BD + AB'C'D'

**OR**

Design a combinational circuit whose input is 3-bit number and whose output is the 2's complement of the input number. 05

Q.3 (a) Implement the Boolean expression F(A,B,C) = $\sum$ m (0,3,5,6) using 4:1 multiplexer. 02

(b) Explain the working of Encoder. 02

(c) Explain full substractor. 03

**OR**

Construct a 5□32 decoder using four 3□8 decoder and one 2□4 decoder 03

(d) Implement full adder with two half adder and one OR gate with its truth table 05

**OR**

Design a Binary to Gray code converter. 05

Q.4 (a) What is the difference between Sequential Circuit and combinational circuit? 02

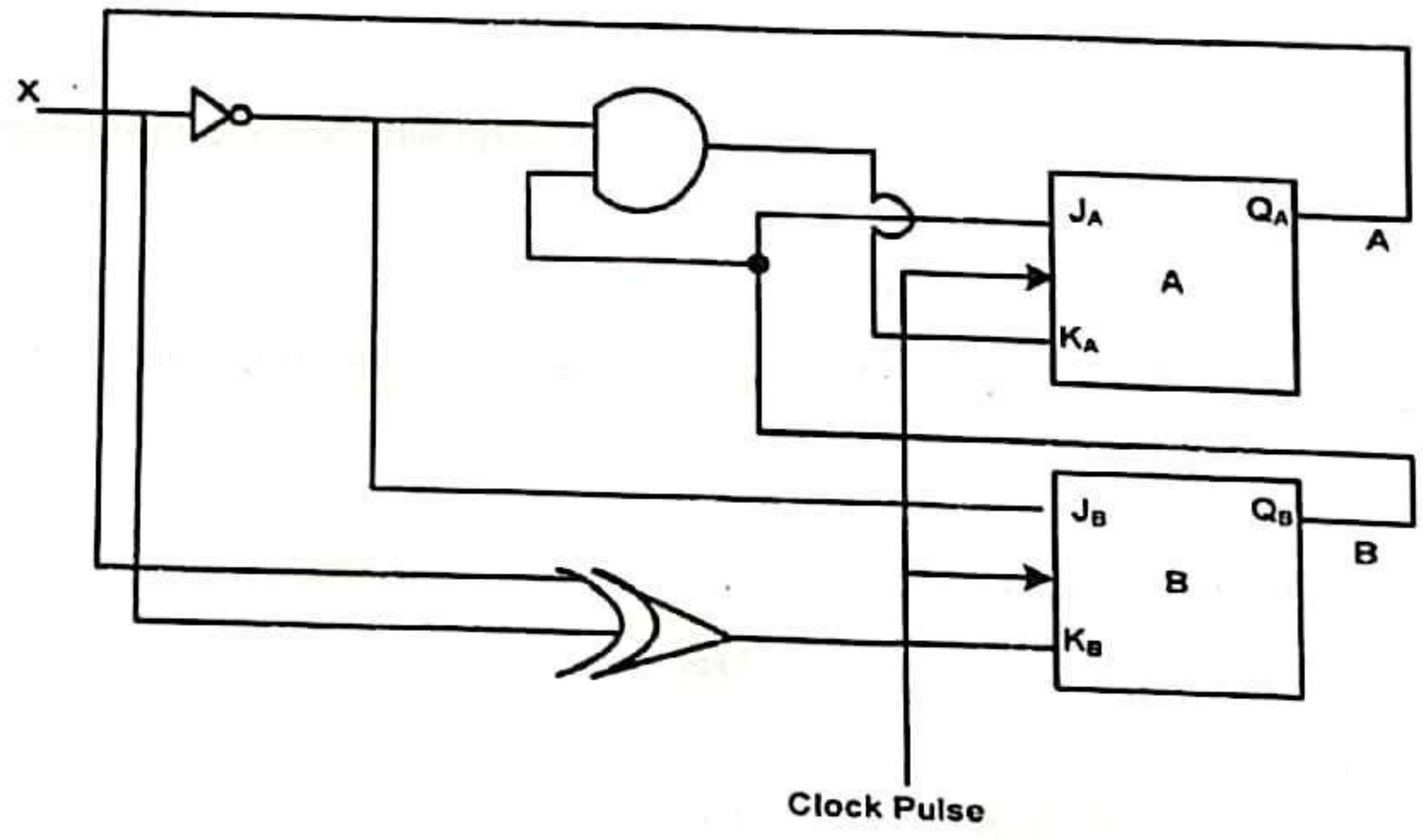
(b) Explain SR Latch flip-flop using NAND Gate. 02

(c) Explain JK flip-flop with characteristics and Excitation table. 03

**OR**

Convert JK Flip Flop to T Flip Flop. 03

(d) Find the state diagram 05




Scanned with OKEN Scanner

*Explain different type of shift register explain them with neat diagram.* 05

Q.5 (a) Write a difference between Asynchronous and Synchronous counter. 02

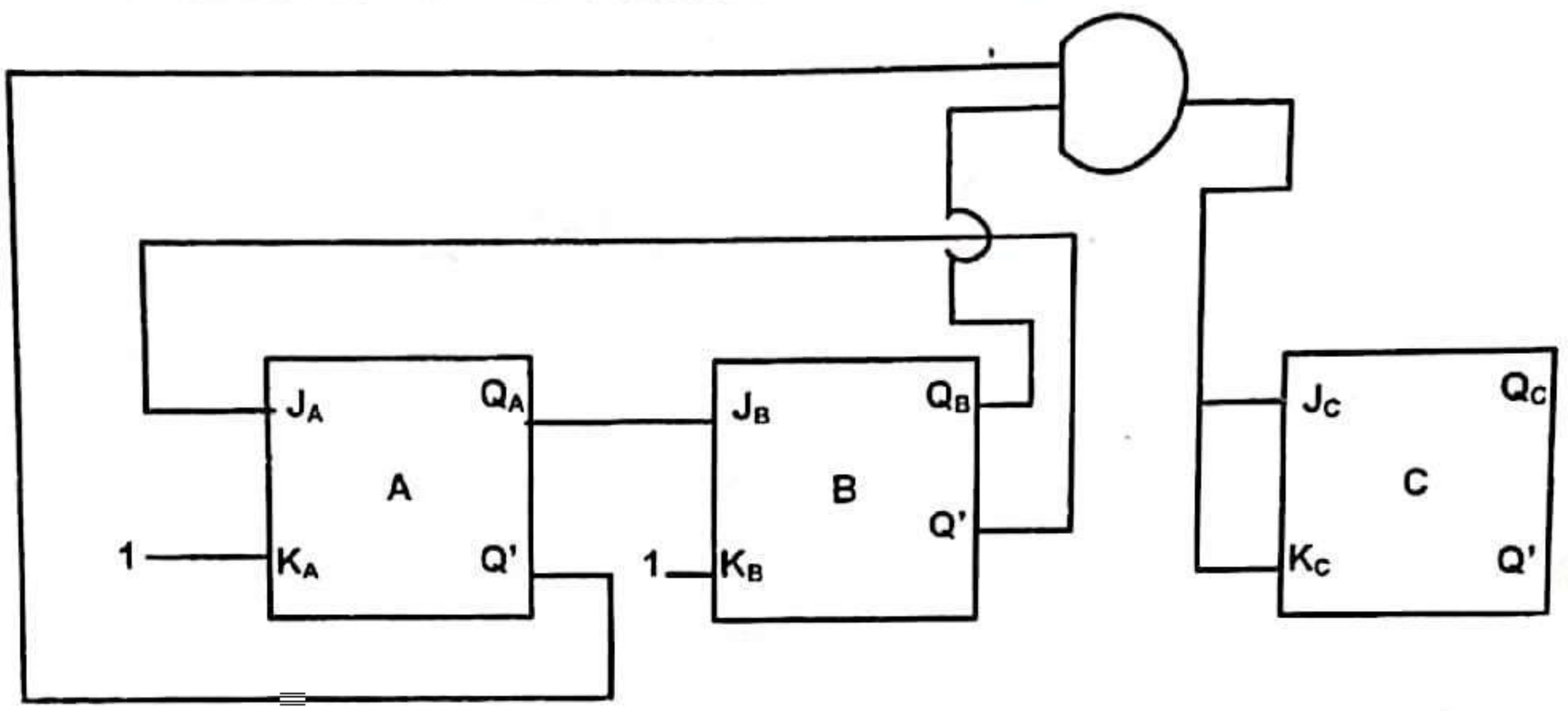
(b) Design 2-bit asynchronous up counter using J-K flip-flop. 02

(c) Design a 3-bit Synchronous up counter using T flip-flop. 03

**OR**

Design a synchronous counter that will count 0 , 3, 5, 7. 03

(d) Calculate the next three state of counter if the initial state is '0 0 0'. 05




**OR**

Design a mode 6 down asynchronous counter for active low pre-set. 05

*****

Scanned with OKEN Scanner